WAKE TECH COMMUNITY COLLEGE

Information Technology Division · Networking Technology

Introduction to Networks

NET-125

Fall 2026 Section 0006 3 Credit(s) 10/15/2026 to 12/16/2026 Delivery Method: Online
Modified 10/18/2026

How & When our Course Meets

Course Format:

Asynchronous Online: Fully online course without scheduled meetings; College curriculum or continuing education course in which 100% of the instruction is delivered online. Students who do not complete the Course Entry Quiz by the 10% mark of class noted on the course calendar will be removed from class and marked as "Never Attended."

Meeting Days / Times / Location:

not applicable

Schedule Adjustment and Withdrawals

Each semester, you have a window of opportunity to make changes to your course schedule or withdraw. The timing of when you withdraw impacts how much, if any, of your tuition may be refunded and may also impact your financial aid.

To view the withdrawal deadlines, please visit Wake Tech's Calendar (https://www.waketech.edu/student-services/registration-student-records).
To view the comprehensive withdrawal policy, visit Withdrawal Policy (https://www.waketech.edu/catalog/academic-information)
To contact financial aid, visit Financial Aid (https://www.waketech.edu/%EF%AC%81nancial-aid)

Meet Your Instructor

Refer to the instructor's information in our Blackboard course > **Faculty Information**.

What This Course Is About

This course introduces the architecture, structure, functions, components, and models of the Internet and other computer networks. Topics include introduction to the principles of IP addressing and fundamentals of Ethernet concepts, media, and operations. Upon completion, students should be able to build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes.

The curriculum for this course will be the Cisco Networking Academy CCNAv7 Introduction to Networks course.

Requisites

CTI-120

What You'll Learn: Student Learning Outcomes

Upon completion of this course, students will be able to:

1. Define network terminology.
2. Explain the OSI and TCP/IP models.
3. Explain how the network, data link, physical, application and transport layer protocols enable and support network connectivity and communication.
4. Explain how devices access local and remote network resources.
5. Describe switch and router hardware.
6. Explain how Ethernet works in a switched network.
7. Calculate numbers between decimal, binary, and hexadecimal systems.
8. Describe IPv4 and IPv6 addressing.
9. Design and implement an IPv4 and IPv6 addressing scheme in a network.
10. Configure initial settings on a switch, router and end user devices.
11. Implement security best practices on switches and routers.
12. Verify and troubleshoot connectivity in a small network.

What You'll Need: Textbook & Materials

The entire curriculum for the NET 125 is available online, through the Internet. Course content for this class is found on the website for the Cisco Networking Academy (NetAcad). Once your instructor has officially entered you as a class member within the NetAcad database, you can log into your homepage at the NetAcad Web Site (Opens in a new window) (https://www.netacad.com/) {plain text link: https://www.netacad.com/ } to view the course content. As such, no textbook is required for this course.

However, some students like a hard copy of the material they are studying. If so, you may purchase an optional textbook through the Wake Tech Bookstore. Here is a link: Wake Tech Bookstore (https://www.waketech.edu/student-services/bookstore) (URL https://www.waketech.edu/student-services/bookstore)

The Companion Guide provides the online content and supplemental material. This text is for students who are seeking a hard copy of the course content and some additional material beyond what is presented in the curriculum. This book has content, pictures, end-of-chapter questions, key terms, glossary, etc. You will need to access the online curriculum for videos and activities.

If you are buying the hard copy textbook from a third party, be sure it is **Version 7 (copyright 2020)** to correctly match the current content taught in the curriculum used in the NET 125 class. Earlier printed versions (prior to 2020) do not match what is taught in class. This textbook is also available electronically for free to Wake Tech Students through O'Reilly for Higher Education (see information below regarding access to O'Reilly). Some students like using this version of the book because they can highlight, etc. The textbook:

Introduction to Networks: Companion Guide (CCNAv7)
ISBN-13: 9780136633662
Cisco Press, 2020

Internet Access:
This class is heavily Internet-based. You will need a high-speed Internet connection to do the various activities expected of you on time.

Required Software:
The NetAcad curriculum will run on Windows and Apple computers. Chromebooks are known to have issues. There are directions in Module 1 in Blackboard for accessing your NetAcad account. Cisco recommends using **Chrome** or **Firefox** when accessing NetAcad. You will need to have pop-ups enabled for the Cisco NetAcad site and cookies enabled. Many homework assignments done in NET 125 require you to use a virtual network simulation software package called Packet Tracer. You will download Packet Tracer from the NetAcad website and install it on your computer. Packet Tracer is available for Windows, macOS, and Linux. Packet Tracer will not run on Chromebooks. Be sure to install the latest version of **Packet Tracer, version 9.0.1 or later**. If you have an earlier version of Packet Tracer, you need the current version. Packet Tracer was updated for v7.02 of the NetAcad curriculum. Some students have indicated that some of the menu items, etc. are in different places in the macOS version. If you use multiple computers, you may install a copy of Packet Tracer on each computer. The **Packet Tracer** software is **free** to **NetAcad students**. Once your instructor has officially entered you as a class member within the NetAcad database, you can set up your NetAcad account and then download Packet Tracer from NetAcad. There are directions in Module 1 of this course for setting up your NetAcad account and accessing the NetAcad Content. In NetAcad Module 1 of our course is a video with details about downloading and installing Packet Tracer. This is found in NetAcad on page **1.0.3 - Downloading and Installing Packet Tracer**. Directions for installing Packet Tracer are also in Blackboard under course resources.

You will download and install Wireshark for a lab assignment. Wireshark is free software. Details for installing Wireshark are included with the lab assignment.

While Packet Tracer is an excellent tool for learning skills, experience using real Cisco equipment increases skills. Students in this class will complete labs in the classroom or in NetLabs using real equipment. NetLabs is real Cisco equipment accessible via the Internet. There is no cost to the student for NetLabs. Many employers favor students that have experience with real equipment.

Other Supplies:
For **seated** classes, you will need to use Wake Tech One Drive to move lab files from one computer to another prior to submitting the lab for grading

Eagle Advantage is Wake Tech's textbook program for students in degree programs, excluding Career & College Promise (CCP) and Cooperative Innovative High School (Early College) students. The program ensures all required printed and digital textbooks are bundled and ready for pickup or delivery on or before the first day of class - saving you time and money and ensuring you're ready to learn on Day One.

If you are opted-in to Eagle Advantage, visit the 'Course Materials-BibliU' link in Blackboard for more information about the textbook details for your course. You can also find out more information by visiting FAQs and Troubleshooting BibliU (https://support.bibliu.com/hc/en-us/sections/360004011038--FAQs-and-Troubleshootin)

If you opted-out of Eagle Advantage, you are responsible for purchasing any of the required texts/materials for this course. For more information about Eagle Advantage, contact our Bookstore (https://www.waketechcampusstore.com/SelectTermDept)

Need Computer Access or Wi-Fi?

> You'll need access to a computer for this course. If you don't have your own computer, there are several options: Visit one of the Wake Tech libraries or open computer labs. Additionally, many public libraries offer free computer access. Want to check out a laptop for a few weeks? Wake Tech can help! Contact the Wake Tech Libraries about the Loaner Device options (opens in new window).

[alternate link: https://www.waketech.edu/student-services/libraries]

✓ How Your Work will be Evaluated: Grading Information

Breakdown

Grading Scale

This course uses a 10-point grading scale.

- A: 90% and up
- B: 80% - 89%
- C: 70% - 79%
- D: 60% - 69%
- F: < 60%

Criteria

Please see final grade details below.

Type	Weight	Topic	Notes
NetAcad Checkpoint Exams	20%	Module Group Exams	
Assignments & Packet Tracer Labs	30%	Packet Tracer Labs	
Skills Final Exam	20%	PT Skills Final Exam	
NetAcad Final Exam	10%	NetAcad Final Exam	
Blackboard Final Exam	20%	Blackboard Final Exam	

How You'll Proceed Through the Course: Course Calendar

The normal course schedule is below. For the **specific semester schedule**, please refer to the **Weekly Schedule** under Course Resources in Blackboard. Also, see Blackboard for a more detailed list of assignments and due dates.

When	Topic	Notes
NetAcad Modules 1 and 5 Week #1	Networking Today, Number Systems	

When	Topic	Notes
NetAcad Modules 2 and 3 Week #2	Basic Switch and End Device Configuration, Protocols and Models	
NetAcad Modules 4, 6, and 7 Week #3	Physical Layer, Datalink Layer, Ethernet Switching	
NetAcad Modules 8 - 10 Week #4	Network Layer, Address Resolution, Basic Router Configuration	
NetAcad Module 11 Week #5	IPv4 Addressing	
NetAcad Modules 12 - 14 Week #6	IPv6 Addressing, ICMP, Transport Layer	
NetAcad Modules 15 - 17 Week #7	Application Layer, Network Security Fundamentals, Build a Small Network	
Final Exam Week #8	NetAcad Final Exam, Blackboard Final Exam, Skills Final Exam	

✳ How I Will Help You Succeed: Instructor Policies & Procedures

Attendance

For in-person format and blended format, the college catalog student handbook states that "students are expected to be in attendance at least 90 percent of all scheduled class meetings. An absence is defined as missing one-third or more of any regularly scheduled class meeting." If you anticipate an absence, notify your instructor in advance. "If prior notification is not possible, the student should contact the instructor immediately upon returning to the College to determine the next course of action." See details in Blackboard Course Resources, Course Policies for details for your section of this course.

Attendance for asynchronous online format is measured by dated submission of assignments, graded assessments in NetAcad or participation in weekly discussion boards. See details in Blackboard Course Resources, Course Policies for details for your section of this course.

Assignment Submission and Due Date

See the Course Resources, Grading Information section in Blackboard for details regarding assignment submission and due dates.

Email

Every student is provided with an official Wake Tech email account through the student portal (my.waketech.edu). Students must first activate their my.waketech.edu account, wait 24 hours, and then activate the email account. This college-issued email account is to be used for all email correspondence with instructors and other college officials. Once the account has been activated, students may choose to re-direct their email to a personal account. However, students are reminded that official correspondence from the college (communications from instructors, information about registration or financial aid, etc.) will ONLY be sent to the Wake Tech email address, and that instructors and college officials may refuse to accept student emails sent from other addresses.

Follow these procedures when trying to reach the instructor:

(1) Email communication between student and faculty must come through your provided Wake Tech email account to ensure compliance with federal privacy requirements.

(2) Expect your instructor to respond to emails within 24 hours.

(3) Be specific about the subject of the email in the mail subject heading and use proper spelling, grammar, and punctuation.

How the Department Will Help You Succeed: Department Policies & Procedures

The Tutoring and Learning Center (TLC) (https://www.waketech.edu/student-services/tutoring-learning-center) offers Wake Tech students free tutoring and learning support for their classes. The TLC's mission is to help students understand what they are learning in class, develop the skills they need to be successful in college and the workplace, and empower them to be independent learners.

Information & Digital Technology Advising Team (https://www.waketech.edu/student-services/advising/information-technology)

The team provides specialized advising and coaching support for University Transfer students interested in IT and Associate in Applied Science students enrolled in the following programs:

- Business Analytics
- Cloud Infrastructure
- Computer Programming and Development
- Cybersecurity
- Data Science and Programming Support Services
- Graphic Design
- IT Service and Support
- Network Management
- Simulation and Game Development
- Web & UX Design
- Web Developer

Each student is assigned to an academic advisor within their care team. To find your advisor, go to Self-Service, click "Student Planning," then "Plan and Schedule," and then the "Advising" tab.

How the Division Will Help You Succeed: Division Policies & Procedures

Process for Expressing Concerns about the Course

1. Please discuss concerns with your instructor first.
2. If your concerns are not resolved, you may contact the Department Head or Program Director of the department.

General Division Questions

For general questions regarding the Information Technology Division or assistance with program resources, students may contact Associate Dean Emmie Mercer at ermercer1@waketech.edu for guidance or referral to the appropriate department.

Information Technology Technical Standards

This page outlines the technical standards and essential skills expected of students in Information Technology programs. Review this page for descriptions of the critical-thinking, communication, interpersonal, behavioral, and technical abilities needed to complete coursework and perform effectively in professional IT environments: IT Technical Standards (https://www.waketech.edu/about-wake-tech/divisions/information-technology/technical-standards)

How the College Will Help You Succeed: Wake Tech Services, Policies and Procedures

Wake Tech is committed to supporting you by providing the resources needed to succeed. This is a brief overview of support available at the College. You can always contact me to help you make the right connection with a college resource. We are here to help!

Student Support

As a Wake Tech student, you have free access to non-academic and academic resources to help you with a range of challenges. These resources include academic services, such as advising, tutoring and academic coaching to help you understand your career and academic pathway, Disability Support Services, Tutoring and Learning Centers, Career Services, and Library Services. The college also offers assistance with basic life needs through the Care Center. These services include mental health resources, emergency financial assistance, safe housing, transportation, and technology. If you find yourself in need, you can reach out to any faculty member or visit this website Student Support (https://www.waketech.edu/studentsupport) to learn more about the help that we can provide.

Student Activities

Wake Tech is continually developing new ways for you to get involved with the College! For information about the Student Centers, Student Government Association, Clubs and Organizations, Athletics, and more, please visit the College Catalog (https://www.waketech.edu/catalog/student-services-and-activities).

Technical Support

Wake Tech has several technology platforms to help you succeed. You may find that technical questions come up regarding your use of technology and Wake Tech is here to help. For technical support, including Blackboard, WTCC email, MS Teams, and Self Service visit the IT Helpdesk page (https://www.waketech.edu/help-center/its/topics).

Student Code of Conduct, Rights, and Responsibilities

You may have questions about your rights and responsibilities as a student. Wake Tech has a range of policies to address such concerns. For information about the Academic Integrity Policy, Complaints, Title IX Policy, Student Appeals, and more, please visit the College Catalog (https://www.waketech.edu/catalog/student-code-conduct-rights-and-responsibilities).

Academic Policies

There are several Wake Tech policies related to academics that you might need to reference in order to succeed in your classes. If you have questions about the Attendance Policy, Student Email Policy, Absences for Religious Observances, Withdrawal Policies, Grades, and more, please visit the College Catalog (https://www.waketech.edu/catalog/academic-information).

Student Information Privacy

Wake Tech is committed to protecting your privacy. The Family Educational Rights and Privacy Act (FERPA) is a federal law enacted in 1974 that protects the privacy of student education records. To find out more about College policies and procedures that support your privacy, please visit the College Catalog (https://www.waketech.edu/catalog/registration-and-student-records).

Campus Policies

There are several Wake Tech policies related to campus procedures that you might need to reference to feel comfortable on campus. For information about College Police and Safety, the Acceptable Use of College Resources Policy, (Student Dress, Pets, Visitors and Children on Campus, and more, please visit the College Catalog (https://www.waketech.edu/catalog/campus-policies-and-procedures).

Registration & Student Records

For information about the Class Schedules, Registration Dates, Transcript Requests, Transfer Credits, Program of Study Changes, and more, please visit the College Catalog (https://www.waketech.edu/catalog/registration-and-student-records).

Financial Aid

Wake Tech is here to answer your financial aid questions. For information about Enrollment Requirements, Grants, Loans, Scholarships, and more, please visit the College Catalog (https://www.waketech.edu/catalog/financial-aid).

Inclement Weather/Wake Tech WARN

North Carolina can have some unpredictable weather. For up-to-date information about Emergency Closings, please sign up for the College alert system, Wake Tech WARN (https://www.waketech.edu/about-wake-tech/administrative-offices/campus-police/waketech-warn). For more information about the inclement weather policy, please visit the College Catalog (https://www.waketech.edu/catalog/campus-policies-and-procedures).

Additional Items

Departmental Mission

The Network Management program prepares individuals for employment supporting local and wide area networks. Students learn how to use technologies to provide data, voice, image, and video communications in business, industry, and education.

Departmental Goals

- **Prepare Workforce-Ready Graduates**
  Equip students with the technical knowledge, hands-on skills, and professional behaviors required for immediate employment in networking and IT support roles.
- **Align Curriculum with Industry Standards**
  Ensure coursework reflects current industry practices and aligns with recognized certifications, including CompTIA Network+, Security+, Cisco CCST/CCNA, and cloud platform credentials.
- **Emphasize Experiential Learning**
  Provide authentic, lab-based learning experiences using enterprise-grade tools, virtual labs, and real-world scenarios that mirror workplace environments.
- **Develop Professional and Soft Skills**
  Foster communication, teamwork, critical thinking, documentation, and customer-service skills essential for success in IT and technical service roles.
- **Support Career Pathways and Transfer Opportunities**
  Create clear pathways to employment, apprenticeships, internships, and transfer to four-year institutions in information technology and related fields.

- Promote Ethical and Responsible Computing
  Instill best practices in security, privacy, accessibility, and ethical decision-making in all aspects of networking and systems administration.
- Encourage Lifelong Learning
  Prepare students to adapt to evolving technologies by cultivating habits of continuous learning, pursuing certifications, and engaging in professional development.
- Advance Equity and Student Success
  Provide an inclusive, supportive learning environment that values diverse backgrounds and ensures all students have access to the resources they need to succeed.

Departmental Leadership Contact Information

Department Head: Kobie Joyner

Email: kkjoyner@waketech.edu

Phone: 919-335-1271

Office Location: RTP1- 306F